

Exploring the Transferability of ULTRA to PICKLE

– Evaluation of a Graph Foundation Model on a Plant Science Knowledge Graph Corpus

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Abstract

This project investigates the transferability of [Galkin et al. \(2024\)](#)’s ULTRA framework to the new dataset PICKLE ([Lotreck et al., 2023](#)) with regard to simplicity and performance. I evaluate the three provided model checkpoints on newly created train, validation, and test sets based on PICKLE and report performance for the zero-shot scenario as well as after one, five, ten, and twenty epochs of fine-tuning for each model instance. These experiments show that there are no critical difficulties in transferring ULTRA to PICKLE, but that performance is relatively low in all configurations.

1 Introduction

These days, the paradigm of pre-training foundation models and fine-tuning them on specific tasks is widely employed for Large Language Models (LLMs). However, foundation models for graphs seemed to be out of reach due to the absence of transferable relation representations until [Galkin et al. \(2024\)](#) proposed ULTRA. Their framework produces relation representations conditioned on their interactions and enable the transfer of models to new datasets with unseen relations. The authors present their approach as a break-through for graph foundation models and emphasise on ULTRA’s transferability to any new Knowledge Graph (KG) dataset.

My project evaluates this claim by transferring ULTRA to a new dataset – PICKLE – from the entirely new domain of molecular plant sciences. I investigate the simplicity as well as the performance of adding a new dataset to the ULTRA family and running zero-shot inference as well as inference after fine-tuning. To ensure comparability to the results [Galkin et al. \(2024\)](#) present, I utilise the same task of Knowledge Graph Reasoning (KGR) as a test bed and report extensive performance results for various model configurations. A true founda-

tion model applicable to various knowledge retrieval tasks would be revolutionary for a variety of distinct fields such as knowledge augmentation for LLMs as well as biological, medical, and legal applications relying on reliable information from large collections of data. The claim that ULTRA enables all that must be thoroughly investigated.

2 ULTRA Model

[Galkin et al. \(2024\)](#) contribute a new framework named ULTRA to pave the way towards foundation models for graphs. Well-known from the areas of language and vision, foundation models are pre-trained on massive amounts of data and can be fine-tuned for various specific tasks. With this paradigm, the computational effort of training a model from scratch is only made once, after which the fine-tuning step can be done on only small amounts of data for a few training epochs – essentially saving on computational resources. For this approach, the models rely on representations consistent between the data they are trained on and those they are developed to process. In the past, transferable representations of relations between nodes were in particular problematic and models were usually not transferable between datasets with different relations. With their ULTRA framework, [Galkin et al. \(2024\)](#) provide a technique to generate relations representations conditioned on their interactions and hence make them transferable to graphs with unseen relations.

Based on the observation that there are certain invariances in the interactions between relations across different knowledge graphs, ULTRA constitutes the following three steps. First, the original knowledge graph is lifted to a graph of relations, where the relations of the former now serve as the nodes in the latter. The edges of the graph of relations are labelled with four basic interactions between the relations. They describe how the re-

lations are positioned with respect to each other in the KG. In the second step, the respective representation matrices are obtained by employing a message-passing GNN on the graph of relations to obtain its edge representations. The edge representations of the graph of relations then correspond to relation representations of the original KG. Together with the original query, these relation representations can then be fed to any off-the-shelf link predictor – another message-passing GNN – to perform knowledge graph reasoning.

Galkin et al. (2024) provide three model checkpoints (3g, 4g, 50g), which are each trained on a different mixture of KGs comprising three, four, or fifty graphs respectively. They compare ULTRA against state-of-the-art baselines on 57 transductive and inductive datasets and find that ULTRA outperforms the supervised SOTA approaches on average in the zero-shot scenario. After fine-tuning ULTRA on the respective datasets, the performance increases further. Within their evaluation, the authors emphasise on the zero-shot transferability to any knowledge graph dataset and prove their statement with comparisons of ULTRA and models trained from scratch on each individual dataset, where (fine-tuned) ULTRA again outperforms most end-to-end models.

3 PICKLE Data

PICKLE – short for the Plant ScIenCe KnowL-edge Graph corpus – is a gold standard corpus for the molecular plant sciences and was developed by Lotreck et al. (2023). They employ techniques originally developed for Natural Language Processing (NLP) and present a collection of 250 abstracts from publications in the area of plant sciences annotated with entities and relations. As Lotreck et al. (2023) state, Named Entity Recognition (NER) and Relation Extraction (RE) are those most interesting to the fields of biological sciences from the broad collection of NLP methods. Exemplary entities include genes as well as proteins, corresponding relationships would then be *activates*, *produces*, or *interacts*. As usual for knowledge graph corpora, the relations between entities are presented as triples as for example (*‘JA-induced AabHLH1’*, *‘activates’*, *‘artemisinin biosynthesis’*).

The authors develop an entity and relation ontologies based on the GENIA corpus term and static relations ontologies (Kim et al., 2003) and the BioNLP13 Gene Network Regulation task re-

lations (Bossy et al., 2015). The relation ontology includes five relations categorised into causal and static relations visualised in Figure 3 in the Appendix, while the entity ontology comprises a more complex hierarchy including chemicals, organisms, and anatomy. These ontologies were applied in the annotation of 250 abstracts of journal articles collected by searching PubMed¹ for the two terms *jasmonic acid* and *gibberellic acid*. Inter-annotator agreement reaches 0.65 for entities and 0.49 (strict) or 0.63 (loose) for relations in their F1 agreement score. Overall, the corpus contains 6245 entity and 2149 relation annotations across the 250 documents. The authors present various experiments proving their dataset’s utility for training NER and RE models achieving comparable performance to other state-of-the-art models in their respective domains.

4 Data Conversion

Lotreck et al. (2023) provide PICKLE in two different formats – brat (Stenetorp et al., 2012) and DyGIE++ (Wadden et al., 2019), of which I utilised the former for reasons of accessibility. Brat-formatted data is stored in text files containing the text of the original documents and annotation files containing different types of annotations. In the case of PICKLE, the annotation files contain entity annotations including an ID, the entity type, the corresponding span of characters in the respective document, and the entity label as well as relation annotations including an ID, the relation label, and two arguments referring to its head and tail entities.

As Galkin et al. (2024) describe in their GitHub repository², they provide two base classes for reading in transductive or inductive datasets and one should implement a new subclass inheriting from the respective base class to run ULTRA on a new dataset. Of these two base classes, the TransductiveDataset requires three links to the training, validation, and test sets in a .txt-file containing one triple of head entity, relation, and tail entity per line. To achieve this, I utilise the Python package mendelai-brat-parser³ to initially read the original brat-formatted PICKLE corpus files. Exploring the format identified the crucial step as unifying the 250 documents and their entity

¹<https://pubmed.ncbi.nlm.nih.gov/>

²<https://github.com/DeepGraphLearning/ULTRA?tab=readme-ov-file#adding-your-own-graph>

³<https://pypi.org/project/mendelai-brat-parser/>

and relation IDs, which were not consistent across documents. For example, the entity ID *TI* could correspond to different entities as well as the entity *Giberellic acid* could be assigned different IDs when looking at different documents. Hence, I discard the original IDs and assign new ones to each entity and relation – now consistently across documents. Relating the arguments of the relations to the correct entities is done via a dictionary specific to each document mapping old to new IDs.

To avoid losing information when storing triples of IDs only, the mappings between IDs and entities or relations are saved in JSON files providing a label for each relation and label as well as description for each entity. This approach is guided by the format [Safavi and Koutra \(2020\)](#) provide their CoDEX datasets in, which are in the set of datasets utilisable with ULTRA already. The triples of (head entity ID, relation ID, tail entity ID) are then shuffled, split into train, test, and validation sets as described in Table 1, and finally written to text files. The produced files are made available on GitHub⁴ to provide the download links required by ULTRA’s dataset module. The Python code corresponding to the above steps is available on GitHub⁵ as well.

Set	# Triples	Split in %
Train	1719	80
Validation	215	10
Test	215	10
Overall	2149	

Table 1: Split of PICKLE for evaluating and fine-tuning ULTRA.

5 Experiments

The experiments described in the following section evaluate the performance of ULTRA models on PICKLE in the zero-shot as well as a fine-tuning scenario with respect to two criteria – simplicity and performance. [Galkin et al. \(2024\)](#) emphasise on both and illustrate how ULTRA is easily transferable to unseen knowledge graphs and can perform on-par with or better than state-of-the-art models

trained from scratch on those datasets. The Python code for running the experiments is again available on GitHub⁶

Simplicity encompasses how well the transfer to a new dataset is documented and how easy it is to implement. As for the documentation, the GitHub repository linked to the original paper provides a designated section explaining how to add one’s own graph with the provided dataset classes and a detailed documentation on how to run the supplied model checkpoints on a chosen dataset as well as how to fine-tune and pre-train new ULTRA instances. The explanations are detailed but not overwhelming and implementing a test run on a dataset distributed with ULTRA can be done smoothly on Google Colab⁷. For running ULTRA on a new dataset, the implementation of the new dataset class is self-explanatory after examining those classes already provided. The only caveat is the format required to be read by the dataset module, which is not documented within the ULTRA repository or paper. However, adopting the format of a datasets already included as guidance as described above with CoDEX ([Safavi and Koutra, 2020](#)) solves this problem. Overall, running ULTRA on a new dataset is in fact as simple as the authors claim it to be.

Performance however, is what determines the utility of transferring ULTRA to new datasets. To thoroughly evaluate this, I report the performance of the three foundation model checkpoints 3g, 4g, and 50g in five training scenarios each, including zero-shot performance and performance after one, five, ten, and twenty epochs of fine-tuning the foundation models on the complete PICKLE training set. For each condition, I report Mean Rank (MR), Mean Reciprocal Rank (MRR), Hits@1, Hits@3, and Hits@10 as these are the metrics ULTRA provides for validation as well as testing in Table 2 in the Appendix. The code for the following visualisation and further figures are also available on GitHub⁸.

Figure 1 reports MRR and Hits@10 on the test set for all model configurations and illustrates the following findings. All model instances show

⁴https://github.com/cora-hai/PICKLE_data-split

⁵https://github.com/cora-hai/knowledge-augmentation-project/blob/main/data_conversion.ipynb

⁶<https://github.com/cora-hai/knowledge-augmentation-project/blob/main/experiments.ipynb>

⁷<https://colab.google/>

⁸https://github.com/cora-hai/knowledge-augmentation-project/blob/main/model_evaluation.ipynb

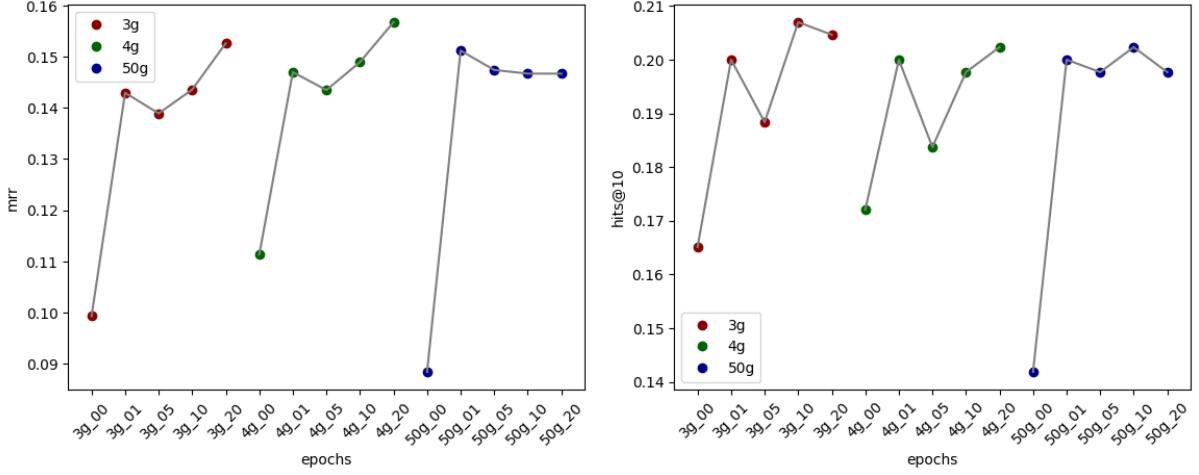


Figure 1: Mean Reciprocal Rank (MRR) and Hits@10 on the test set for each model configuration.

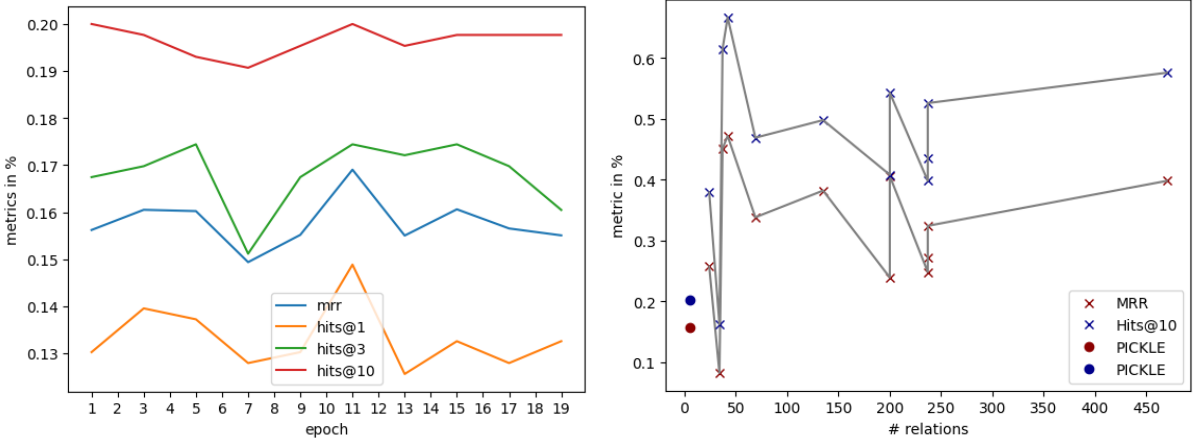


Figure 2: Right: MRR, Hits@1, Hits@3, and Hits@10 on the validation data during fine-tuning of the 4g model for 20 epochs. Left: Plotting the number of relations in each dataset vs. MRR and Hits@10 to investigate a possible correlation between number of relations and performance.

an overall low performance with MRR between 0.0884 for the zero-shot scenario with the 50g model and 0.1568 for the highest-performing instance, which is the 4g model fine-tuned for 20 epochs. Confirming [Galkin et al. \(2024\)](#)’s claim that fine-tuning increases performance, the figures for both MRR and Hits@10 show visible increases after one epoch of fine-tuning for all model sizes. Strikingly, performance decreases after five epochs and – while MRR increases again for 10 and 20 epochs for 3g and 4g – longer fine-tuning does not lead to improvements in MRR for the larger 50g model. Hits@10 does not show a clear trend for a positive impact of fine-tuning for more than one epoch either. The model instance with the best performance averaged over all metrics except MR is the 4g model fine-tuned for 20 epochs.

This partly matches Figure 4 in the Appendix,

which illustrates the development of the loss on the training set during fine-tuning of the 4g and 50g foundation models. Consistent across all instances, the loss decreases steadily for the first three epochs and then begins to oscillate while showing a persistent decreasing trend overall. The lowest training loss is reached in the last epoch for both model sizes. While this seems promising in itself, it is only partly consistent with the development of the performance metrics on the test set, especially for the 50g model, which reaches its highest MRR after only one fine-tuning epoch.

This inconsistency is captured by Figure 2a as well showing the development of the metrics on the validation data during fine-tuning of the 4g model for 20 epochs. Mean Rank was excluded for readability reasons as it uses a different scale than all other metrics. All presented metrics resem-

ble the oscillations of the training loss and illustrate how the performance does not significantly increase with more epochs.

While the low performance can be caused by various reasons, the most striking difference to all other datasets from the original paper is the small set of relations in PICKLE. Figure 2b plots the number of relations in vs. the performance of fine-tuned ULTRA on the given transductive datasets to investigate a possible correlation between the number of relations and ULTRA performance. The performance of ULTRA on PICKLE is marked separately from the other models. The plot indicates no clear relationship between the number of relations and MRR nor Hits@10. However, the dataset with the second-least relations contains 24 relations (vs. five in PICKLE) and the statistic comprises only 12 datasets other than PICKLE. Investigating the influence of the number of different relations on model performance may hence be a promising direction of future work.

6 Conclusion

What this project demonstrates is that applying the ULTRA pipeline to a new knowledge graph dataset is as simple as Galkin et al. (2024) claim, but does not necessarily result in high performance on knowledge graph reasoning. Thoroughly investigating the reasons for ULTRA’s overall low performance on PICKLE is not covered in the scope of this project. A first step would be comparing the ULTRA instance with the best performance on PICKLE to a model trained from scratch on the dataset to evaluate if the quality of the data itself produces low performance. A step in the same direction would be to raise the PICKLE entity labels to broader categories based on the established ontology. If performance increases, one may argue that the number of different entities is too high for the size of the dataset and that not enough training examples for each type exist. Furthermore, expanding the exploration of the correlation between the number of relations and performance could reveal a certain threshold in the number of relations licensing an approach based on the interactions between them. If the number of relations in PICKLE is below that threshold, the relation representations may be too similar to each other, which would then explain the rather low performance. Overall, ULTRA is certainly an innovative and simple approach to foundation models for graphs but will

presumably profit from further investigations of the effects of different domains, entity and relation sets, or dataset sizes on model performance before employing it in broader scientific fields.

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A PICKLE Relation Ontology

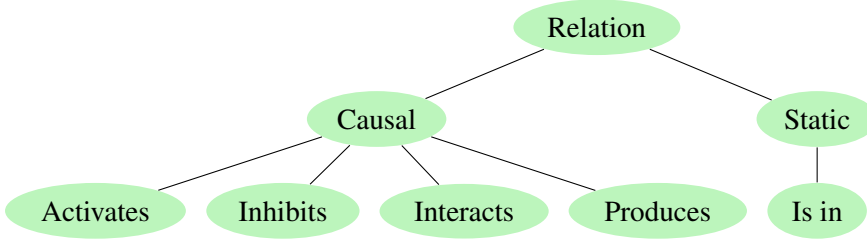


Figure 3: Relation ontology for PICKLE with causal and static relations.

B Full Evaluation Results

	# Epochs	MR	MRR	Hits@1	Hits@3	Hits@10
3g	0	762.5950	0.0993	0.0581	0.1349	0.1651
	1	547.3370	0.1430	0.1070	0.1628	0.2000
	5	540.8020	0.1389	0.1047	0.1605	0.1884
	10	539.2330	0.1436	0.1000	0.1674	0.2070
	20	547.6440	0.1527	0.1209	0.1674	0.2047
4g	0	621.6980	0.1113	0.0628	0.1512	0.1721
	1	540.4000	0.1470	0.1140	0.1628	0.2000
	5	525.8980	0.1435	0.1116	0.1605	0.1837
	10	540.8370	0.1490	0.1233	0.1628	0.1977
	20	546.2300	0.1568	0.1326	0.1651	0.2023
50g	0	640.9090	0.0884	0.0558	0.1093	0.1419
	1	556.6330	0.1512	0.1209	0.1674	0.2000
	5	535.2740	0.1474	0.1163	0.1558	0.1977
	10	541.6120	0.1467	0.1116	0.1558	0.2023
	20	561.4720	0.1467	0.1140	0.1558	0.1977

Table 2: Full evaluation results on three model sizes (3g, 4g, 50g) reporting Mean Rank (MR), Mean Reciprocal Rank(MRR), Hits@1, Hits@3, and Hits@10 for each scenario (zero-shot or fine-tuning for 1, 5, 10, or 20 epochs).

C Training Loss Curves

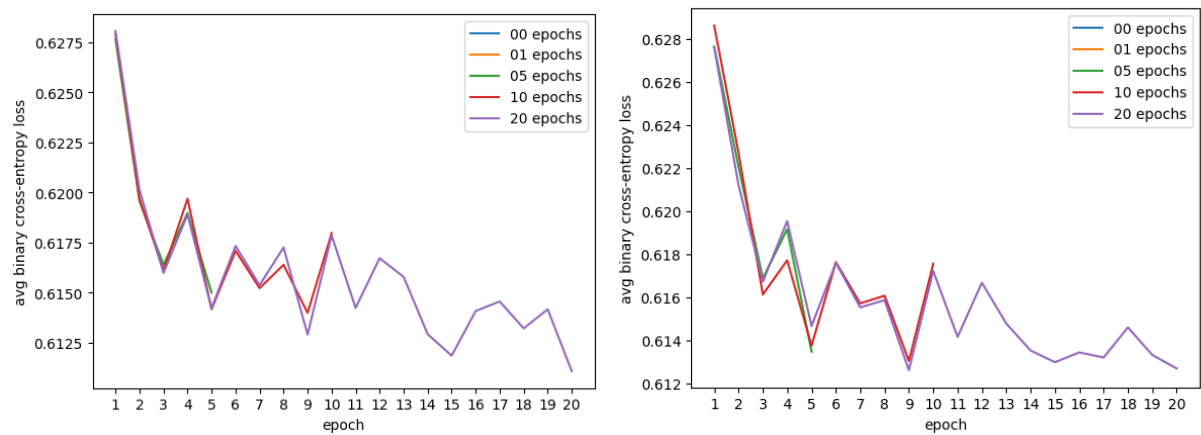


Figure 4: Development of the average binary cross-entropy loss on the training set during fine-tuning of the 4g (left) and 50g (right) foundation models.